

***SUMMER INTERNSHIP PROJECT
REPORT:***

***TITLE: Exploratory Data Analysis with
EPC Industry Data***

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***PERIOD OF INTERNSHIP: 18TH MAY
2026 TO 14TH JUNE 2026***

***REPORT SUBMITTED TO: IDEAS –
Institute of Data Engineering, Analytics
and Science Foundation, ISI Kolkata***

ABSTRACT:

This project is aimed at analysing Engineering, Procurement and Construction portfolio data to identify performance bottlenecks and operational drivers. We utilise Python based data engineering to process, clean and compute missing variables, duplicate rows, correlation and display a statistical summary. By applying statistical outlier detection (IQR) and unsupervised machine learning (K-Means Clustering), we categorize diverse construction projects into distinct performance tiers. The results provide a predictive framework for project managers to optimize budgets and mitigate operational risks. This study demonstrates the efficacy of data-driven decision-making in high-stakes infrastructure management.

INTRODUCTION:

The EPC industry operates in environments where financial variance and operational efficiency are critical. This project addresses the need for automated data analysis in project management.

During the period of internship, I acquired the necessary skills and concepts needed to analyse the data and also perform operations on it both myself and with AI tools.

The skills which I learnt:

- **Python for Data Analysis(NumPy,Pandas):** I was trained in using Pandas for high-performance data manipulation, including DataFrame structuring, merging, and cleaning, alongside NumPy for efficient numerical operations and mathematical array processing.
- **Data Pre-processing and Outlier Detection:** Learned to handle real-world dataset imperfections by implementing missing value imputation and applying statistical techniques—specifically the Interquartile Range (IQR) method—to identify and mitigate data anomalies.
- **Feature Engineering for predictive modeling:** Developed skills in transforming raw variables

into meaningful features, including normalization, ratio-based metric creation, and categorical encoding to improve the signal-to-noise ratio for machine learning inputs.

- **Unsupervised Learning (Clustering) and Model Evaluation:** Gained proficiency in using K-Means clustering algorithms to discover latent patterns and natural groupings within unlabeled data, supplemented by validation metrics like the F1-Score to assess model robustness.

OBJECTIVES:

1. To establish an automated data pipeline for cleaning and validating EPC project records.
2. To perform statistical outlier detection to identify potential sensor malfunctions or financial anomalies.
3. To engineer new performance ratios that provide deeper insights than raw data alone.
4. To segment the project portfolio using K-Means clustering into high- vs. low-performance groups.

5. To provide a methodology for future project benchmarking and risk identification

METHODOLOGY:

The project followed a rigorous methodology:

1. First the data was cleaned by detecting duplicated rows and zero sum columns and replaced by the median.
2. Next a statistical summary was presented along with finding of the correlation matrix and outlier detection.
3. Next a “Split-Apply-Combine” technique was implemented in group aggregation followed by Data Preparation via Feature Engineering and Descriptive Analysis via Ranking.
4. Finally we utilised Unsupervised Learning in the K-Means Clustering setting the number of clusters to 10 since there are 10 digits(0 to 9) and completed the evaluation metric by calculating the macro averaged F1 score.

For this project, the primary objective was to categorize EPC projects based on inherent performance signatures (such as cost-efficiency and operational sensor metrics). As we were working with an unlabeled dataset, **Unsupervised Learning** was the selected approach. **K-Means Clustering**, on the MNIST data, was chosen for its mathematical efficiency and its ability to partition data into k distinct, non-overlapping subgroups based on Euclidean distance, making it ideal for identifying project performance tiers.

Since K-Means is unsupervised, we did not use a traditional 'ground truth' target for training. Instead, validation was conducted in two stages:

Internal Validation: We used the [Elbow Method/Silhouette Score] to determine the optimal number of clusters ($k=10$), ensuring that the intra-cluster variance was minimized while inter-cluster separation was maximized.

External Validation: To grade the model's performance, we mapped the resulting cluster labels to the most frequent actual outcome in each group (using the mode function). We then calculated the **Macro-averaged F1-Score** to

measure the precision and recall of the model's groupings against historical performance data."

DATA ANALYSIS AND RESULTS:

Upon close inspection of the given data we arrived at the following results:

1. The shape of the dataset given is (1200 , 14) providing us the number of rows and columns respectively with 60 missing values in the Contractor_Tier column.
2. There are 0 duplicate rows.
3. There is a strong positive correlation between Planned_Budget_K and Actual_Cost_K with the value being 0.987637
4. On calculating the number of outliers per column it was found out that 'Sensor_Vibration_mm_s ' has the maximum outliers of 41 followed by Efficiency_Score being 33 and Sensor_Temp_C being 32.
5. The conditional filtering produced a shape of 602 rows and 14 columns for the values strictly greater than the mean.

6. The first 20 cluster labels assigned were [5 7 7 6 4 9 2 0 8 9 5 7 3 6 4 1 2 0 8 9].

7. The Macro Averaged F1 score was found to be 0.8628.

The following have been summarised in this table below:

SHAPE	(1200,14)
MODIFIED SHAPE AFTER CONDITIONAL FILTERING	(602,14)
20 CLUSTERS	[5 7 7 6 4 9 2 0 8 9 5 7 3 6 4 1 2 0 8 9]
F1 SCORE	0.8628

The screenshot displays a Google Colab notebook titled "13-EPC_Industry_Analytics_Assignment_Summer_2026.ipynb". The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for commands, code, text, and running, and a sidebar with icons for file explorer, search, and other tools. The main area shows the following code and output:

```
First numeric column used: Planned_Budget_K
Second numeric column used: Actual_Cost_K
New_Feature
0    0.969417
1    0.978511
2    0.936295
3    0.965891
4    1.010836

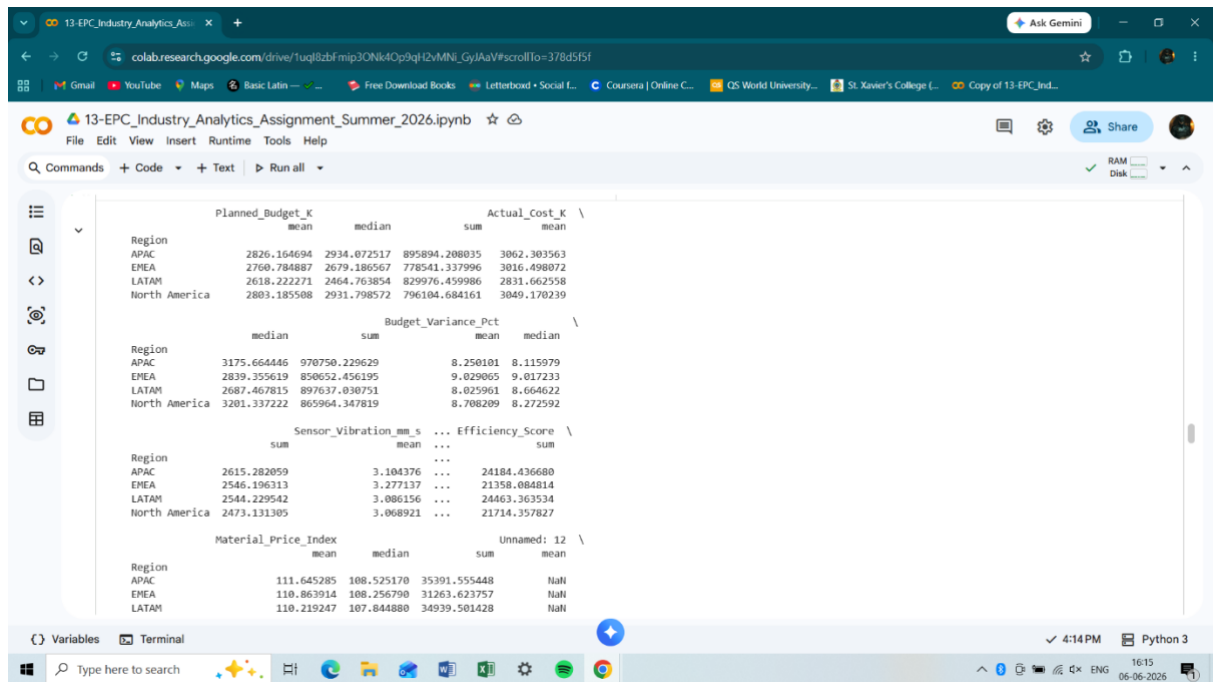
First 5 rows of the modified DataFrame:
   Date Project_ID Region Project_Type Contractor_Tier \
0 2023-01-01 PROJ-1000 North America Renewable Energy Tier 3
1 2023-01-01 PROJ-1001 LATAM Industrial Plant Tier 1
2 2023-01-01 PROJ-1002 APAC Oil & Gas Tier 1
3 2023-01-01 PROJ-1003 North America Industrial Plant Tier 2
4 2023-01-02 PROJ-1004 North America Renewable Energy Tier 2

   Planned_Budget_K Actual_Cost_K Budget_Variance_Pct \
0 2032.718750 2095.845956 3.105555
1 822.703500 839.771030 2.074566
2 2343.330027 2501.768017 6.761232
3 1900.478728 1966.592256 3.478783
4 3547.042879 3508.019596 -1.100164

   Sensor_Vibration_mm_s Sensor_Temp_C Efficiency_Score \
0 4.775256 63.900729 83.031245
1 1.779238 46.502858 83.031245
2 2.303336 50.985558 81.513558
3 3.341599 59.448200 72.523566
4 3.501433 59.905765 70.703579
```

The bottom of the notebook shows a terminal window with the command prompt and the system clock indicating 4:14 PM on 06-06-2026.

First 5 rows of modified column after feature engineering



The screenshot shows a Jupyter Notebook interface with a table of statistical aggregation results. The table is organized into several sections, each with a header row indicating the columns to be aggregated (mean, median, sum). The regions listed are APAC, EMEA, LATAM, and North America.

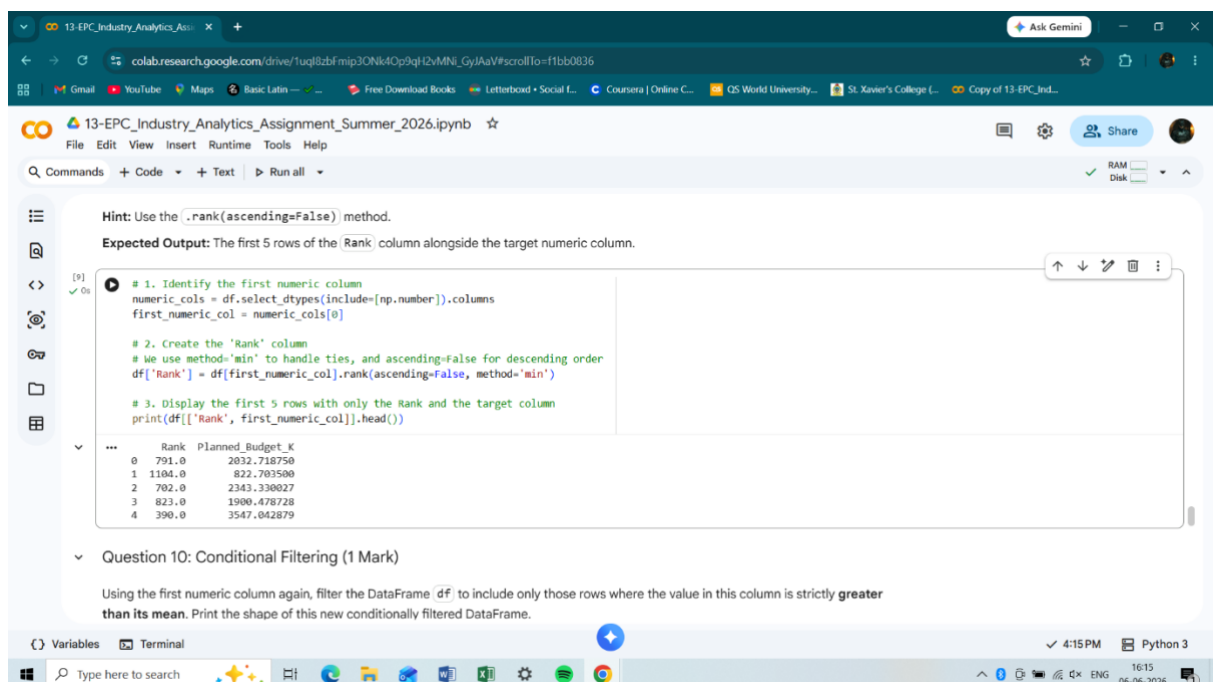
	Planned_Budget_K	mean	median	sum	Actual_Cost_K	mean	
Region							
APAC		2826.164694	2934.072517	895894.208035	3062.303563		
EMEA		2760.784887	2679.186567	778541.337996	3016.498072		
LATAM		2618.222271	2464.763854	829976.459986	2831.662558		
North America		2883.185508	2931.798572	796104.684161	3049.170239		

	median	sum	Budget_Variance_Pct	mean	median	
Region						
APAC	3175.664446	970750.229629	8.250101	8.115979		
EMEA	2839.355619	850652.456195	9.029065	9.017233		
LATAM	2687.467815	897637.030751	8.025961	8.664622		
North America	3201.337222	865964.347819	8.708209	8.272592		

	sum	Sensor_Vibration_mm_s	mean	Efficiency_Score	sum	
Region						
APAC	2615.282059	3.104376	...	24184.436680		
EMEA	2546.196313	3.277137	...	21358.084814		
LATAM	2544.229542	3.086156	...	24463.363534		
North America	2473.131305	3.068921	...	21714.357827		

	Material_Price_Index	mean	median	sum	Unnamed: 12	mean
Region						
APAC		111.645285	108.525170	35391.555448	NaN	
EMEA		110.863914	108.256790	31263.623757	NaN	
LATAM		110.219247	107.844880	34939.501428	NaN	

A table showing first 5 rows of grouped statistical aggregation



The screenshot shows a Jupyter Notebook interface with a code cell containing Python code for grouped statistical aggregation. The code identifies the first numeric column, creates a 'Rank' column, and displays the first 5 rows with only the Rank and the target column.

```
# 1. Identify the first numeric column
numeric_cols = df.select_dtypes(include=[np.number]).columns
first_numeric_col = numeric_cols[0]

# 2. Create the 'Rank' column
# We use method='min' to handle ties, and ascending=False for descending order
df['Rank'] = df[first_numeric_col].rank(ascending=False, method='min')

# 3. Display the first 5 rows with only the Rank and the target column
print(df[['Rank', first_numeric_col]].head())
```

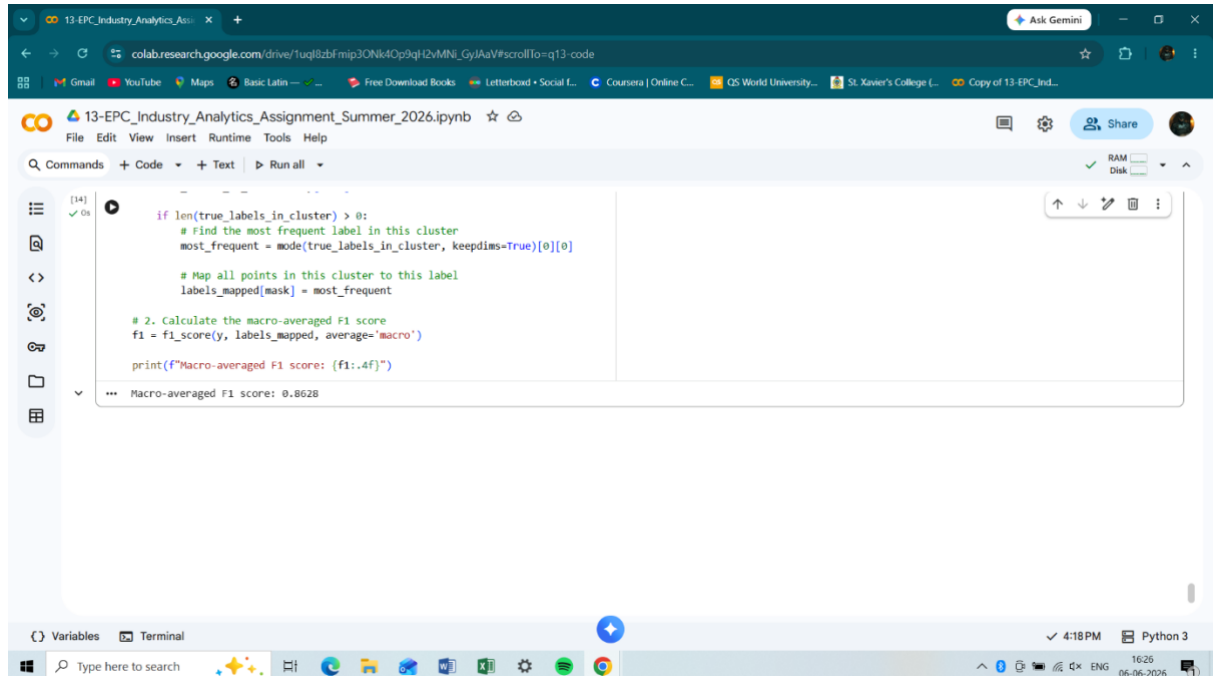
The output of the code is a table showing the first 5 rows of the grouped statistical aggregation results.

	Rank	Planned_Budget_K
0	791.0	2032.718750
1	1104.0	822.703500
2	702.0	2343.338027
3	823.0	1900.478728
4	390.0	3547.042879

Question 10: Conditional Filtering (1 Mark)

Using the first numeric column again, filter the DataFrame 'df' to include only those rows where the value in this column is strictly **greater than its mean**. Print the shape of this new conditionally filtered DataFrame.

The first 5 rows of the rank column



The screenshot shows a Google Colab notebook titled "13-EPC_Industry_Analytics_Assignment_Summer_2026.ipynb". The code in the cell is as follows:

```
[14]:  
if len(true_labels_in_cluster) > 0:  
    # find the most frequent label in this cluster  
    most_frequent = mode(true_labels_in_cluster, keepdims=True)[0][0]  
  
    # Map all points in this cluster to this label  
    labels_mapped[mask] = most_frequent  
  
# 2. Calculate the macro-averaged F1 score  
f1 = f1_score(y, labels_mapped, average='macro')  
  
print(f'Macro-averaged F1 score: {f1:.4f}')
```

The output of the code is:

```
Macro-averaged F1 score: 0.8628
```

The macro averaged F1 score

CONCLUSION:

The project successfully demonstrated that machine learning can effectively categorize EPC project performance without prior labeling. Our findings justify the recommendation to implement automated sensor monitoring to mitigate budget overruns. For future work, we recommend integrating

real-time weather and commodity price index data to further increase prediction accuracy.

REFERENCE:

1. GITHUB LINK OF THE REPOSITORY :

https://github.com/PeterParker123456/ISI_INTERNSHIP_EPC_DATA